Ningbo TengLi Testing Co., Ltd

2nd floor, Block B, Ningbo Testing and Certification Base, No. 66
Qingyi Road, Ningbo National Hi-Tech Zone, Ningbo, Zhejiang
Tel: 86574-8783 6802
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LM-79-19 Test Report

For

Torshare Ltd.

(Brand Name: TORSHARE)

5th Minying Road No.3 Industrial Area Shuitian Shiyan Town Bao'an District, Shenzhen, Guangdong, 518108

Model name(s): TS-LHB5-2FT225W-H-TCP[Blank;(MS)]

Report Type:	Testing and Report According to IES LM-79-2019
Type of Luminaire :	High Bay Luminaires (Commercial and Industrial)
Report Date:	2024-03-08
Prepared By:	Ningbo TengLi Testing Co., Ltd 2nd floor, Block B, Ningbo Testing and Certification Base, No. 66 Qingyi Road, Ningbo National Hi-Tech Zone, Ningbo, Zhejiang

Test & Report By:

Ferrum Li

Engineer: Ferrum Li

Review By:

Garman Mo

Manager: Garman Mo

Note: 1. The results contained in this report pertain only to the tested samples

2. This report does not imply product certification, approval, or endorsement by A2LA, or any agency of the Federal Government.

1.1 Product Information:		
Model Number	TS-LHB5-2FT225W-H-TCP[Blank;(MS)]	
Remark	The "[Blank;(MS)]" represents for motion sensor, it can be (MS)=motion sensor or blank=no motion sensor.	
Representative (Tested) Model	TS-LHB5-2FT225W-H-TCP(0%,4000K) TS-LHB5-2FT225W-H-TCP(100%,5000K)	
Model Difference	N/A	
SKU (if available)	N/A	
Type of Luminaire (for integral lamps, list base type and lamp type)	High Bay Luminaires (Commercial and Industrial)	
LED Manufacturer	Bridgelux Inc.	
LED Model	BXEN-27E-13H-9C1 BXEN-50E-13H-9C1	
Dimming	Continuous	
Integral Controls	Yes	
Sample Number	JDE240301-A1	
Date of Receipt	Mar.06, 2024	
Luminaire Aperture (for downlights)	--	in.
Luminaire Length	--	mm
Luminaires Width	--	mm
Number of Units (modular products)	N/A	s

1.2 Rated Values:	
Rated Voltage / Frequency	120-277Vac, 50/60Hz
Nominal Power	120W/165W/225W(Power adjustable)
Rated Initial Lamp Lumen	--
Declared CCT	4000K,5000K(Color Tunable)

1.3 Test Specifications:

Test item	1. Total Luminous Flux 2. Luminous Distribution Intensity 3. Luminous Efficacy 4. Correlated Color Temperature 5. Color Rendering Index 6. Chromaticity Coordinate 7. Electrical Parameters
Reference Standard	1. IES LM-79-2019 Optical and Electrical Measurements of Solid-State Lighting Products 2. ANSI C78.377-2008 Specifications for the Chromaticity of Solid State Lighting Products 3. CIE 13.3-1995 Method of Measuring and Specifying Colour Rendering Properties of Light Sources 4. CIE 15-2004 Technical Report Colorimetry 5. IESNA LM-16-93 Practical Guide to Colorimetry of Light Source

1.4 Test Methods

1) Photometric and Light Distribution Measurement – Goniophotometer Method: Photometric parameters were measured using the goniophotometer and software. The ambient temperature shall be maintained at 25 ℃ ±1 ℃, measured at a point not more than 1 m from the sample and at the same height as the sample. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Luminous flux, luminaire efficacy, zonal lumen were calculated from the software taken at 1 °vertical intervals and 22.5 °horizontal intervals.
2) Chromaticity Measurement – Sphere-Spectroradiometer Method: Chromaticity parameters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 ºC ±1 ºC. The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral power distribution taken at 5 nm intervals over the range of 380 to 780 nm.
3) Electrical Measurements: Electrical parameters were measured using power meters incorporated in goniophotometer or sphere-spectroradiometer system. The ambient temperature surrounding the sample was maintained at 25 ºC ±1 ºC. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Voltage, frequency, current, power, power factor and total harmonic distortion were measured by and read from the power meter.

2.1 Summary of Test Result

Criteria Item	Measured Value		Compliance	Requirement (DLC V5.1)	
Minimum Total Luminous	29624		Pass	>=10000(-10%)	
Minimum Luminous Efficacy	142.95		Pass	Standard: >= 120(-3%)	Premium: >= 135(-3%)
Minimum Power Factor	0.9477		Pass	>= 0.9(-3%)	
Maximum THD %	11.24		Pass	<= 20(+5)	
Minimum CRI	83.4		Pass	>= 70(-1)	
Minimum R9	9		Pass	>= -40(-1)	
Minimum Rg	95		Pass	>= 89(-1)	
Minimum Rf	83		Pass	>= 70(-1)	
Rcs, h1%	-13		Pass	-18%-23%(-1%)	
CCT (K)	4000K	4213	Pass	<=6500K	
	5000K	5053			
Zonal Lumen Requirement	20-50°	58.6	Pass	>=30(-10)	
Corrected UGR	27.4		Pass	Premium: <28.0	

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2.2 Electrical, Photometric and Chromaticity Measurements

Test date	2024-03-07	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-LHB5-2FT225W-H-TCP(0%,4000K)	**Total Operating Time(min)**	75

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE240301-A1	120.0	60	1.742	208.7	0.9983	3.82
	277.0	60	0.7760	203.7	0.9477	11.24

Photometric Measurement – Goniophotometer Method(Test Distance: 26.00m):

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	29828	29624
Luminous Efficacy (lm/W)	142.95	145.46
Zonal lumens in the 20-50 °	58.6	--
Beam Angle (°)	93.2	--
UGR Viewed Crosswise	27.4	--
UGR Viewed Endwise	27.2	--
Center Beam Candle Power (cd)	13089	--

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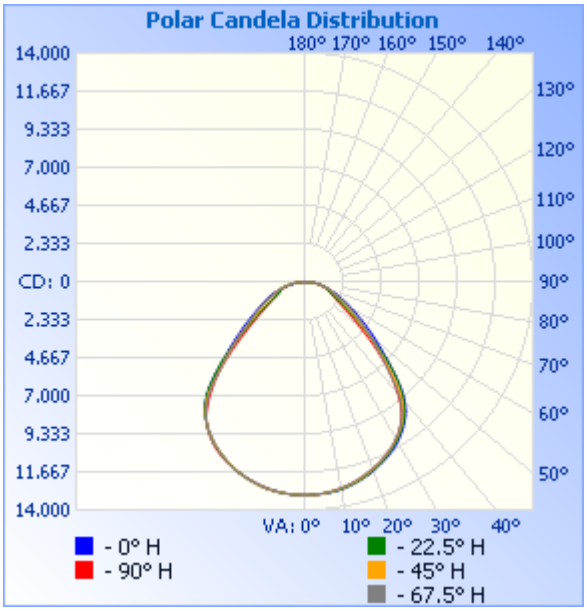
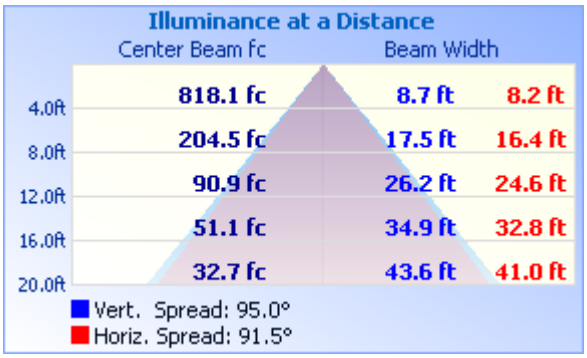
Zonal Lumen Tabulation

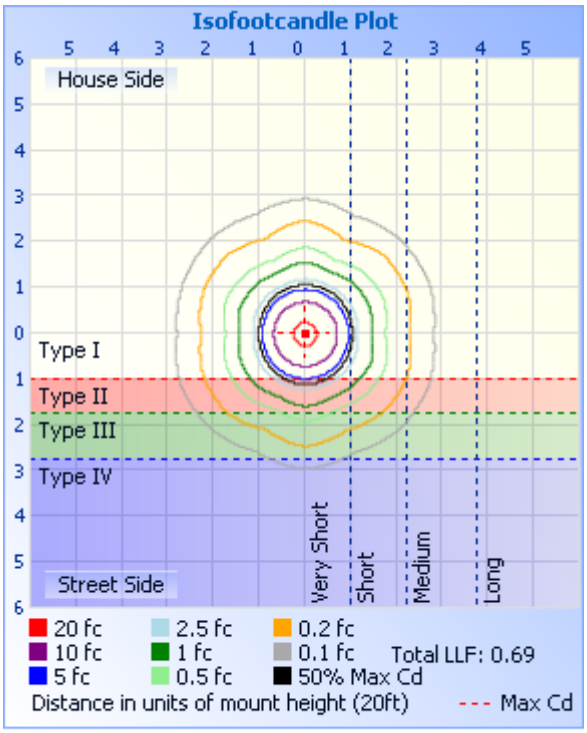
Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	10,249.5	34.4%
0-40	16,753.9	56.2%
0-60	25,739.1	86.3%
60-90	3,996.4	13.4%
70-100	1,798.0	6%
90-120	21.7	0.1%
0-90	29,735.5	99.7%
90-180	89.8	0.3%
0-180	29,825.3	100%

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	%Total
0-10	1,239.3	4.2%	90-100	4.1	0%
10-20	3,564.7	12.0%	100-110	6.7	0%
20-30	5,445.5	18.3%	110-120	10.9	0%
30-40	6,504.4	21.8%	120-130	16.2	0.1%
40-50	5,529.5	18.5%	130-140	16.9	0.1%
50-60	3,455.8	11.6%	140-150	14.5	0%
60-70	2,202.4	7.4%	150-160	11.1	0%
70-80	1,390.7	4.7%	160-170	6.7	0%
80-90	403.3	1.4%	170-180	2.7	0%

Photometric Data







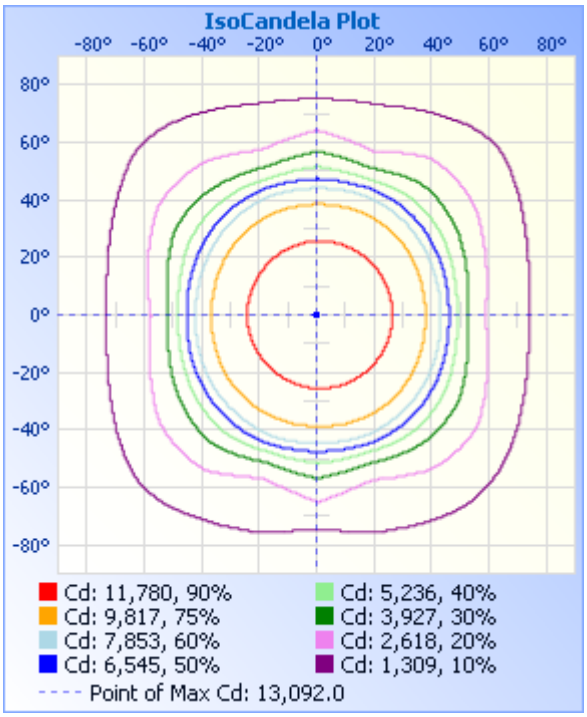


Table--1 UNIT: X10cd

γ (DEG) \ C (DEG)	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5
0	1309	1309	1309	1309	1309	1309	1309	1309	1309	1309	1309	1309	1309	1309	1309	1309
5	1303	1304	1303	1304	1304	1304	1303	1303	1303	1302	1301	1302	1304	1303	1303	1303
10	1287	1288	1290	1293	1292	1292	1288	1284	1285	1283	1285	1288	1291	1291	1291	1289
15	1263	1264	1267	1268	1268	1266	1261	1256	1258	1256	1259	1260	1264	1264	1266	1265
20	1230	1231	1232	1232	1233	1230	1224	1219	1220	1217	1218	1223	1228	1229	1229	1232
25	1189	1188	1188	1190	1190	1184	1177	1174	1175	1172	1168	1175	1181	1183	1182	1186
30	1132	1132	1134	1137	1136	1128	1119	1116	1118	1114	1110	1116	1123	1127	1125	1129
35	1048	1051	1057	1064	1063	1056	1044	1035	1034	1033	1033	1041	1050	1052	1049	1049
40	900	906	932	949	953	944	922	892	886	894	915	935	942	943	930	914
45	694	692	719	758	760	758	713	686	685	685	696	749	742	752	710	696
50	463	518	539	526	582	532	526	530	466	500	523	504	560	510	530	523
55	321	390	411	359	446	365	389	413	320	376	392	338	424	347	385	398
60	236	302	319	251	337	258	299	319	234	297	297	233	323	243	295	316
65	189	234	242	185	255	193	231	239	187	232	237	171	257	181	235	242
70	156	186	183	151	189	155	180	180	155	181	182	151	196	158	186	179
75	120	143	134	130	137	131	136	138	119	136	134	140	124	140	136	126
80	70.6	90.3	91.2	89.2	94.5	92.3	92.3	95.9	77.5	88.2	93.0	87.6	68.5	88.7	90.7	85.1
85	19.6	25.6	35.4	37.8	47.9	40.9	42.3	38.4	32.2	36.1	43.5	34.0	39.5	35.0	34.3	26.2
90	0.39	0.41	0.64	0.55	1.76	0.50	0.41	0.35	0.10	0.14	0.15	0.45	0.19	0.18	0.39	0.13
95	0.40	0.43	0.53	0.58	0.92	0.52	0.44	0.36	0.12	0.15	0.17	0.21	0.20	0.17	0.23	0.14
100	0.48	0.61	0.84	0.78	0.93	0.62	0.61	0.45	0.14	0.17	0.24	0.24	0.26	0.21	0.23	0.20
105	0.72	0.92	1.25	1.06	1.17	0.77	0.82	0.67	0.24	0.29	0.34	0.31	0.36	0.34	0.35	0.37
110	1.07	1.36	1.70	1.64	1.63	1.11	1.01	0.95	0.34	0.39	0.43	0.40	0.50	0.44	0.46	0.48
115	1.59	1.86	2.38	1.52	1.82	1.03	1.34	1.19	0.46	0.53	0.56	0.52	0.63	0.61	0.63	0.63
120	2.25	2.46	3.24	2.08	2.20	1.79	1.77	1.53	0.56	0.70	0.81	0.67	0.80	0.75	0.81	0.74
125	2.95	3.13	3.53	3.20	3.17	2.08	1.92	1.89	0.68	0.90	0.83	0.92	1.06	1.04	0.90	0.97
130	3.64	3.68	3.50	3.48	3.33	2.30	1.94	2.22	1.07	1.05	0.93	1.24	1.48	1.50	1.16	1.12
135	4.00	4.06	2.87	3.46	3.35	2.48	1.54	2.38	1.24	1.15	1.21	1.41	1.76	1.86	1.43	1.26
140	4.15	4.10	1.91	3.36	2.59	2.47	1.19	2.39	1.52	1.44	1.27	1.72	1.76	2.03	1.59	1.52
145	4.22	4.02	2.60	3.32	2.07	2.31	1.72	2.44	1.76	1.65	1.36	1.92	1.76	2.23	1.82	1.74
150	4.23	3.66	3.56	2.90	2.03	2.14	2.73	2.45	1.90	1.72	1.74	1.92	1.92	2.04	2.07	1.88
155	4.03	3.37	4.40	2.63	1.67	1.95	3.14	2.21	1.73	1.63	1.88	1.92	1.83	1.99	2.22	2.10
160	3.77	3.10	4.77	2.66	1.29	1.91	3.16	2.17	1.64	1.58	1.73	1.61	1.75	1.90	2.19	1.89
165	3.49	3.19	4.88	2.89	1.24	2.07	3.18	2.41	1.44	1.38	1.51	1.88	1.72	1.66	1.91	1.82
170	3.36	3.51	5.11	3.07	1.38	2.18	3.41	2.74	1.97	1.90	2.62	3.73	1.99	1.53	2.23	2.86
175	3.28	3.68	5.10	2.73	1.41	1.67	3.40	2.84	2.78	2.75	3.38	4.34	1.91	1.43	1.92	3.11
180	2.73	3.32	4.43	2.10	1.31	1.26	2.92	2.76	2.71	2.75	3.38	4.49	2.16	1.32	1.31	2.99

2.3 Electrical, Photometric and Chromaticity Measurements

Test date	2024-03-07	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-LHB5-2FT225W-H-TCP(0%,4000K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE240301-A1	120.0	60	1.745	209.1	0.9985	3.79
	277.0	60	0.7769	204.0	0.9479	11.21

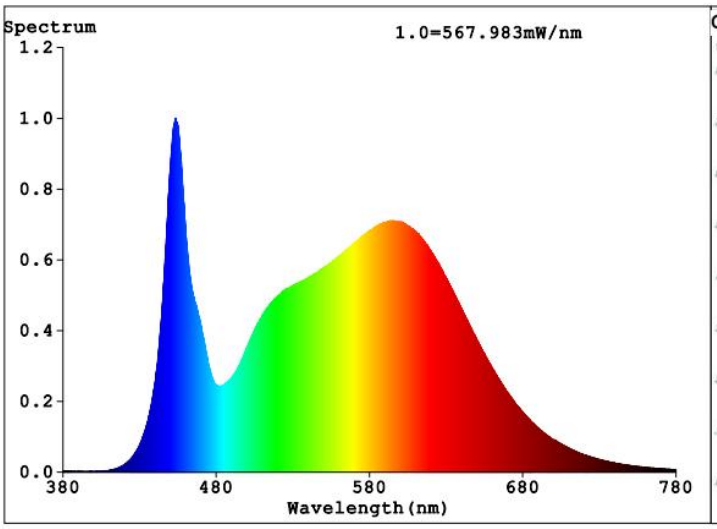
Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.1330)(4π geometry):

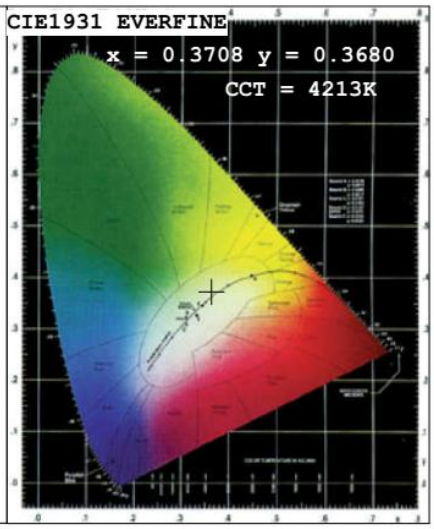
Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	4213
Duv	-0.0012
Chromaticity (x, y)	x=0.3708 y=0.3680
Chromaticity (u’, v’)	u'=0.2222 v'=0.4962
Color Rendering Index (CRI)	85.6
R9	21
Rg	96
Rf	85
Rcs,h1%	-11

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	29941	29739
Luminous Efficacy (lm/W)	143.19	145.78

Spectral Power Distribution & Chromaticity Diagram





R1 =85	R2 =92	R3 =96	R4 =84	R5 =84	R6 =88	R7 =87	
R8 =69	R9 =21	R10=81	R11=83	R12=61	R13=87	R14=98	R15=80

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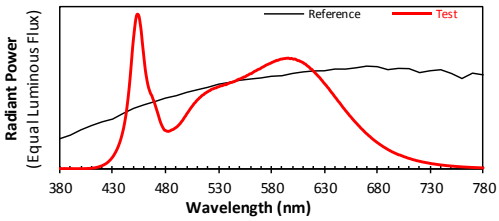
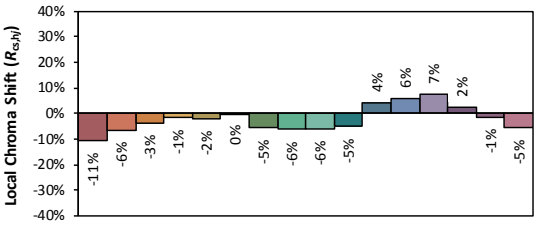
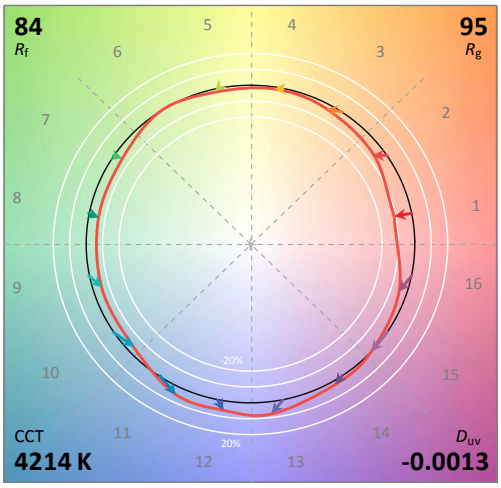
TM30

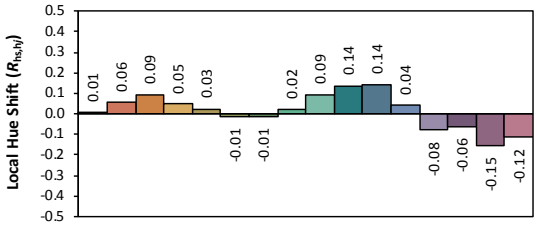
ANSI/IES TM-30-18 Color Rendition Report

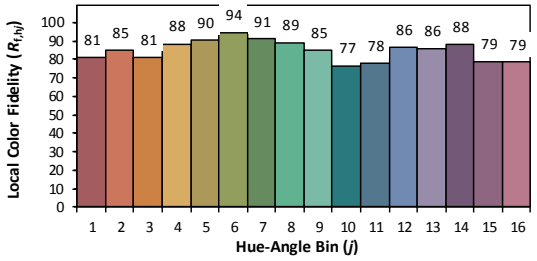
Source:	BXEN-27E-13H-9C1 BXEN-50E-13H-9C1	Manufacturer:	Torshare Ltd.
Date:	2024-03-07	Model:	TS-LHB5-2FT225W-H-TCP(0%, 4000K)

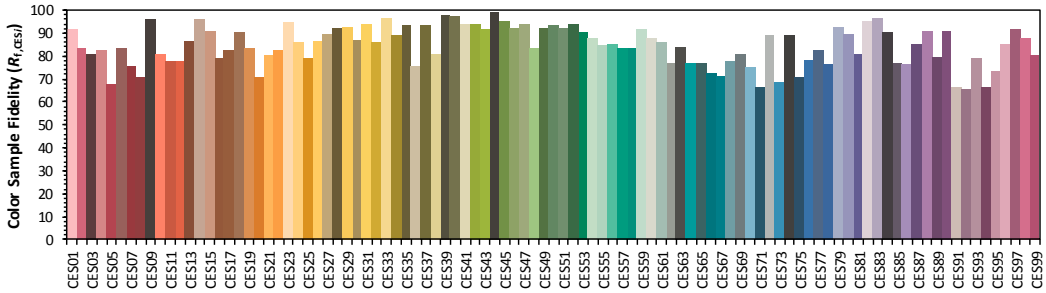












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3707
y	0.3679
u'	0.2222
v'	0.4961

CIE 13.3-1995 (CRI)

R_a	86
R_9	21

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

2.4 Electrical, Photometric and Chromaticity Measurements

Test date	2024-03-07	**Test Ambient:**	25±1 °C
Test Orientation	As intended	**Stabilization Time (min)**	60
Model Number	TS-LHB5-2FT225W-H-TCP(100% ,5000K)	**Total Operating Time(min)**	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE240301-A1	120.0	60	1.739	208.5	0.9991	3.77
	277.0	60	0.7736	203.4	0.9492	11.18

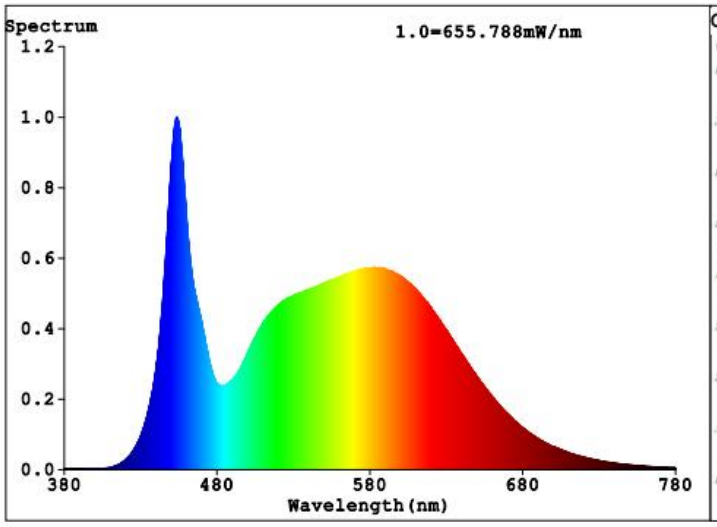
Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.1332)(4π geometry):

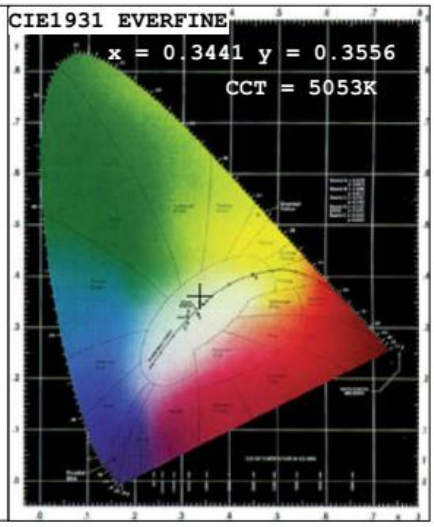
Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	5053
Duv	0.0024
Chromaticity (x, y)	x=0.3441 y=0.3556
Chromaticity (u’, v’)	u'=0.2092 v'=0.4865
Color Rendering Index (CRI)	83.4
R9	9
Rg	95
Rf	83
Rcs,h1%	-13

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	30512	30306
Luminous Efficacy (lm/W)	146.34	149.00

Spectral Power Distribution & Chromaticity Diagram





R1 =82	R2 =90	R3 =94	R4 =82	R5 =82	R6 =85	R7 =87	
R8 =67	R9 =9	R10=74	R11=81	R12=60	R13=84	R14=97	R15=77

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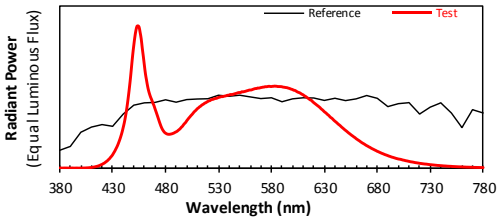
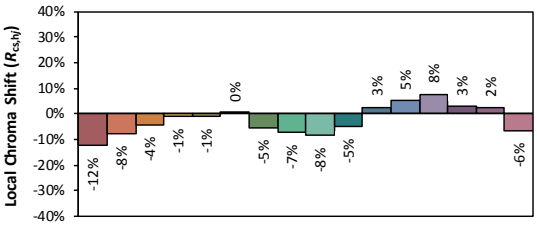
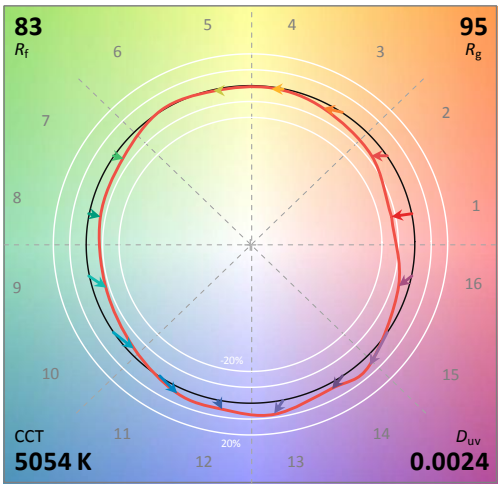
TM30

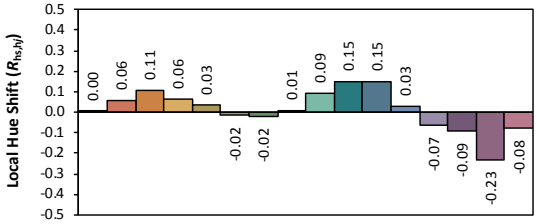
ANSI/IES TM-30-18 Color Rendition Report

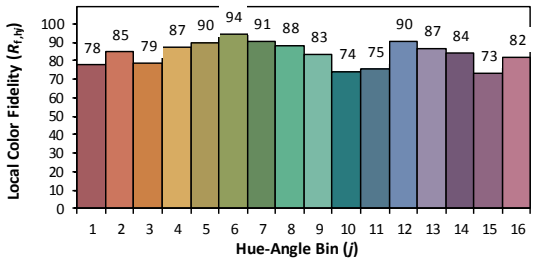
Source:	BXEN-50E-13H-9C1	Manufacturer:	Torshare Ltd.
Date:	2024-03-07	Model:	TS-LHB5-2FT225W-H-TCP(100%,5000K)

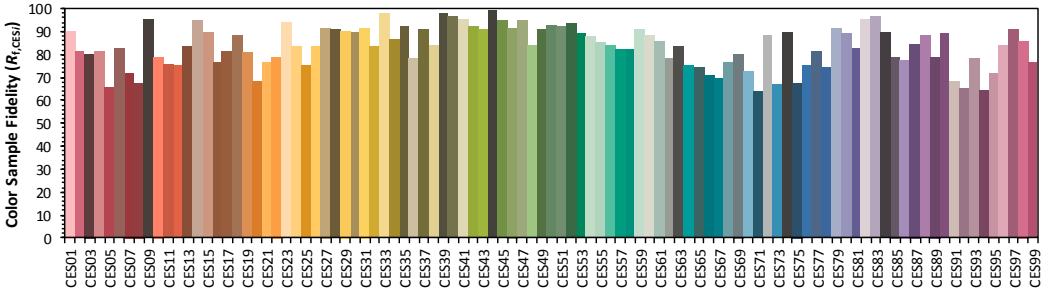












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3440
y	0.3555
u'	0.2092
v'	0.4864

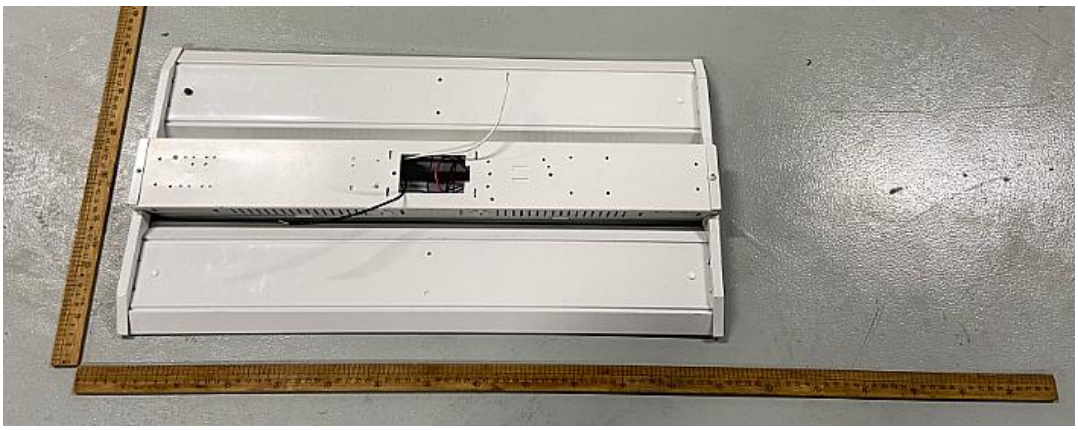
CIE 13.3-1995 (CRI)	
R_a	83
R_9	9

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

3. Test Equipment

Equipment ID	Equipment Name	Last Calibration Date	Next Calibration Date
ST-R-702	2 meter Integrating Sphere	Verified by D204 standard lamp	
ST-R-701	Spectral analysis system HAAS-1200	Verified by D204 standard lamp	
ST-R-703	Standard Lamp D204	2023-12-26	2024-12-25
ST-R-704	Power Meter for Integrating Sphere	2023-12-26	2024-12-25
ST-R-707	Temperature Probe for Integrating Sphere	2023-12-26	2024-12-25
ST-R-714	Goniophotometer system	Verified by D908S standard lamp	
ST-R-710	Standard Lamp D908S	2023-12-26	2024-12-25
ST-R-711	Power Meter for Goniophotometer	2023-12-26	2024-12-25
ST-R-709	Hygrothermograph for Goniophotometer	2023-12-26	2024-12-25
Uncertainty(K=2): Photometric Measurement (Sphere):3.40% Chromaticity Measurement(Sphere):44.8K Photometric Measurement(Goniophotometer):3.64%			

4. Product Photo



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